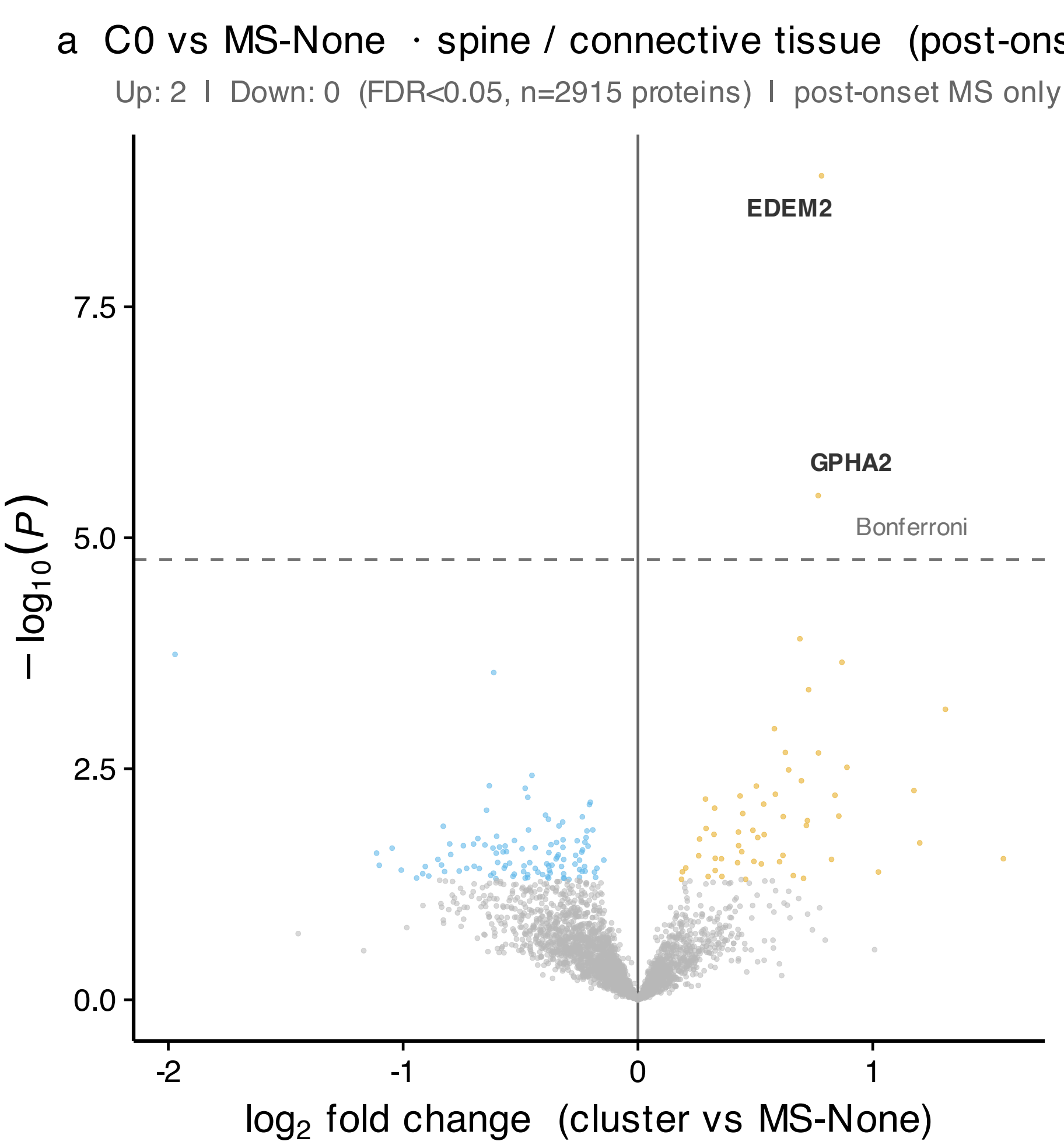
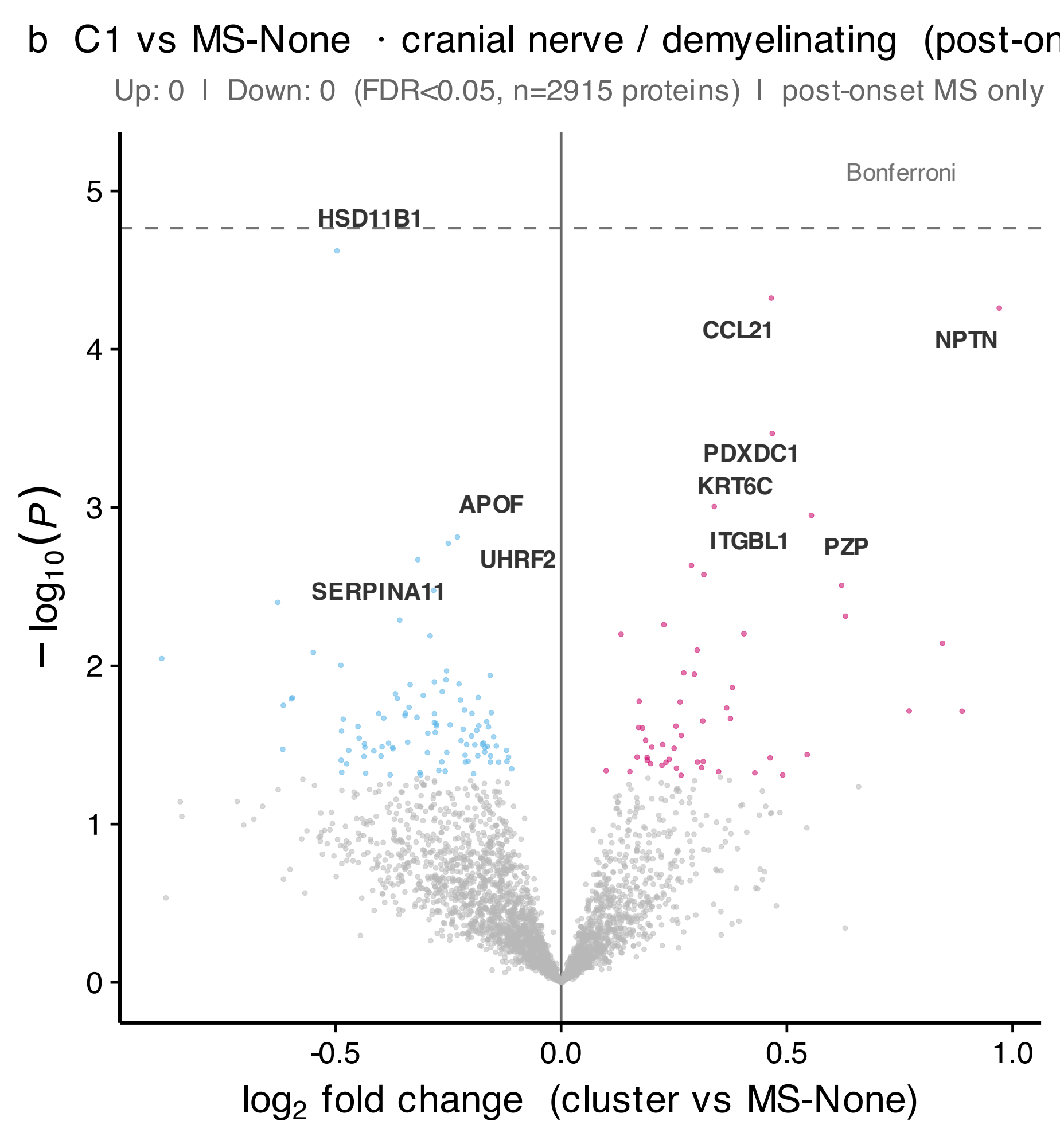
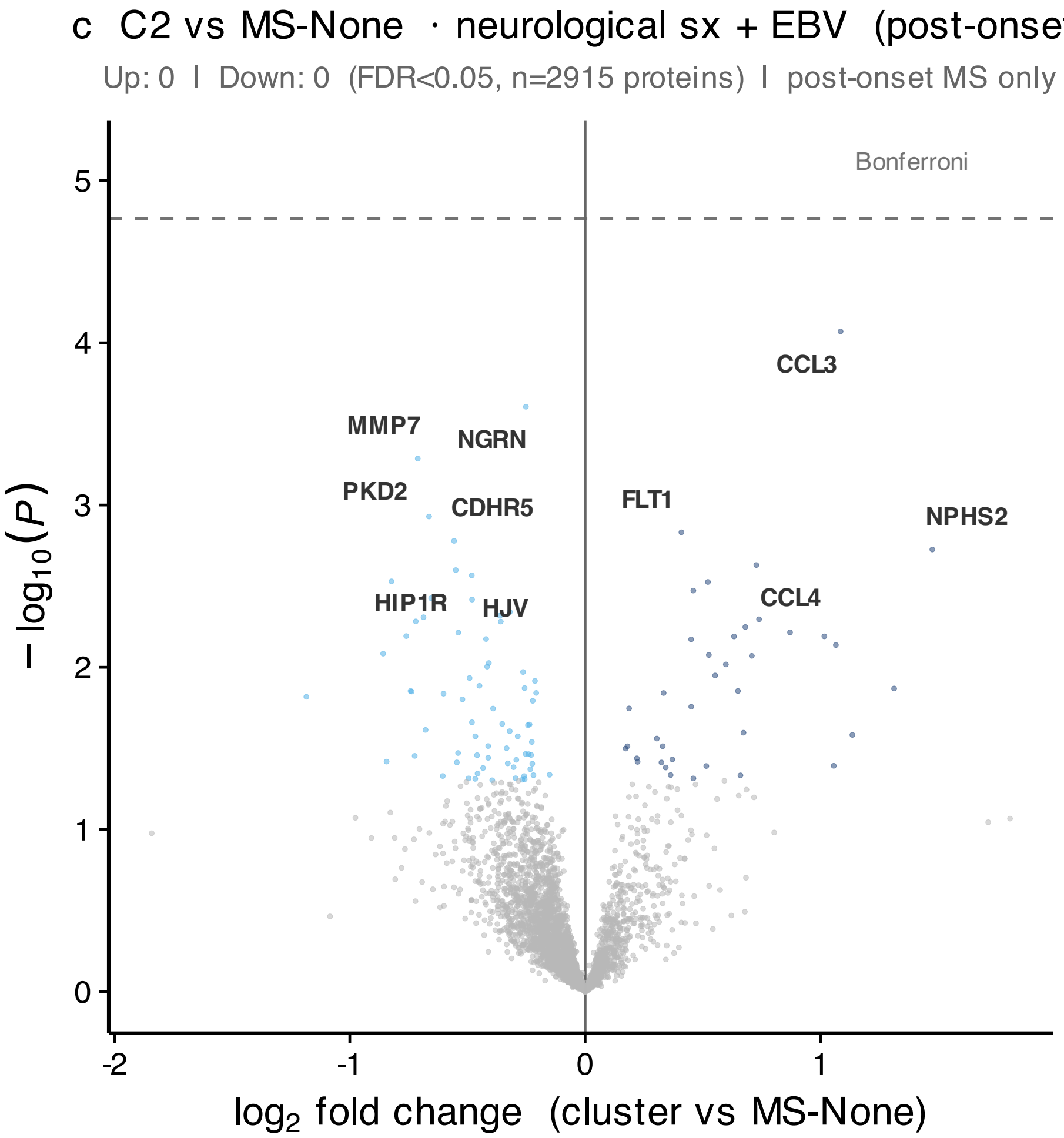
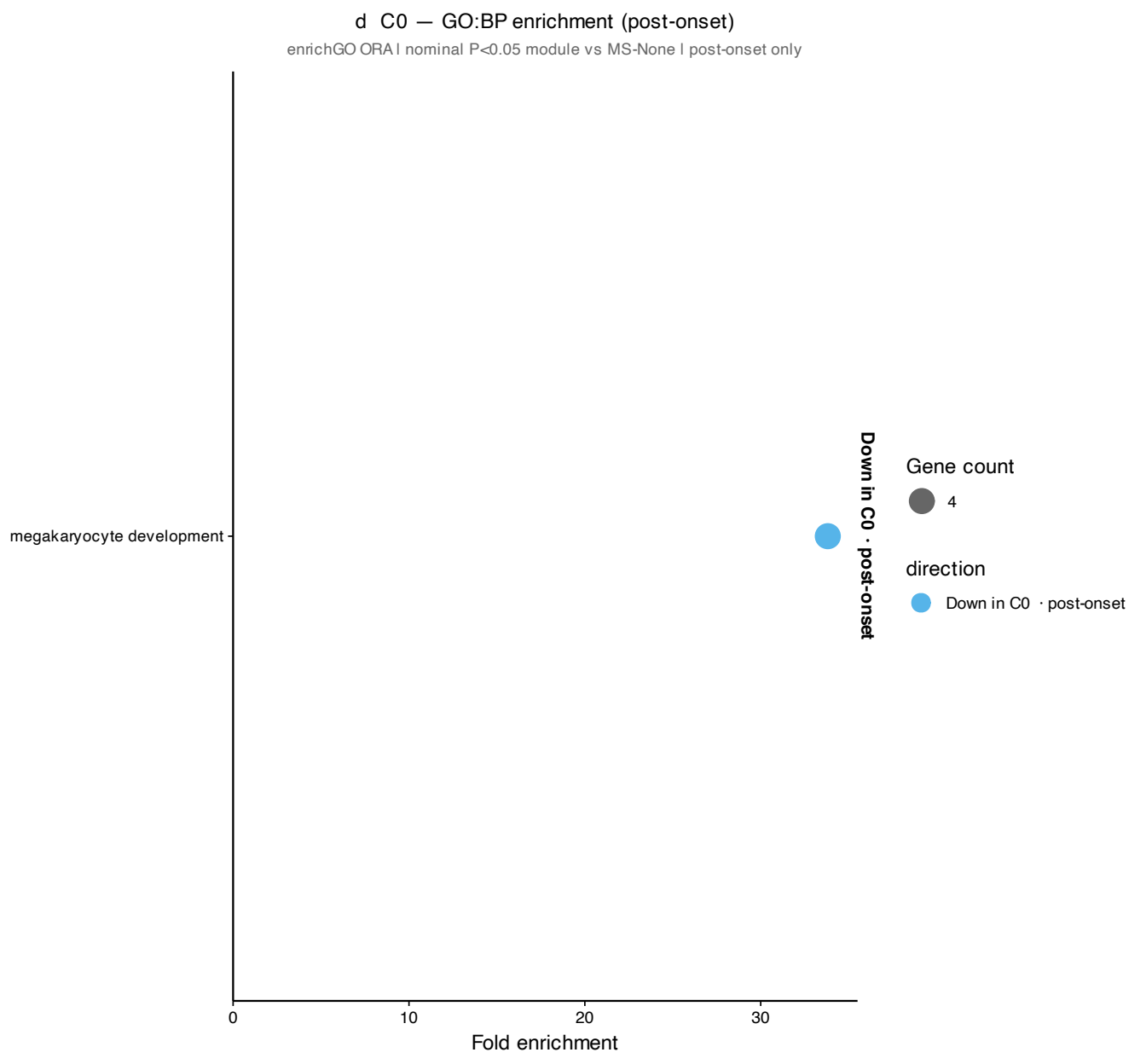
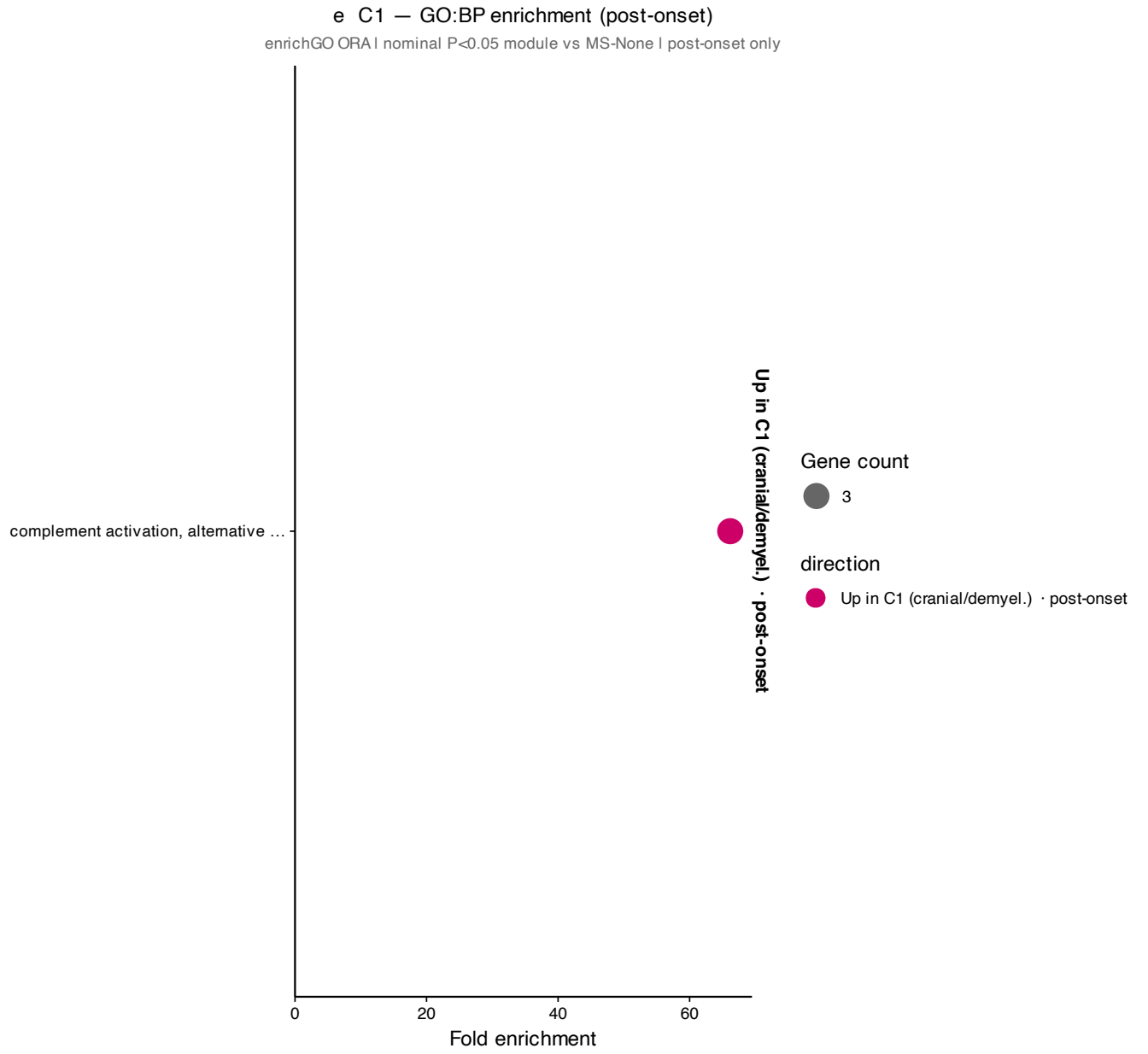
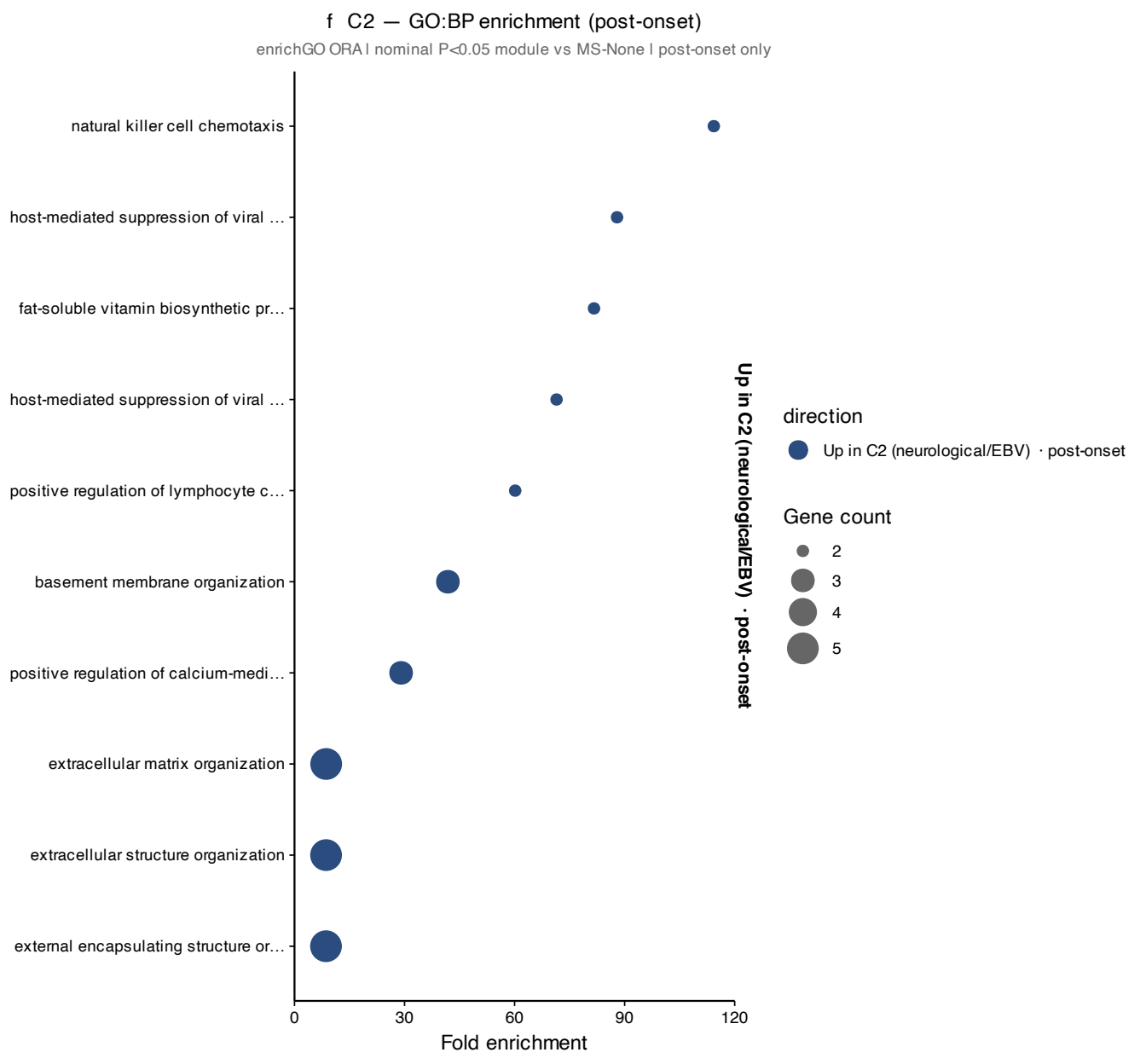

a C0 vs MS-None · spine / connective tissue (post-ons
Up: 2 | Down: 0 (FDR<0.05, n=2915 proteins) | post-onset MS only
EDEM2
GPHA2
Bonferroni
7.5
5.0
2.5
0.0
−log10(P)
-2
-1
0
1
log2 fold change (cluster vs MS-None)

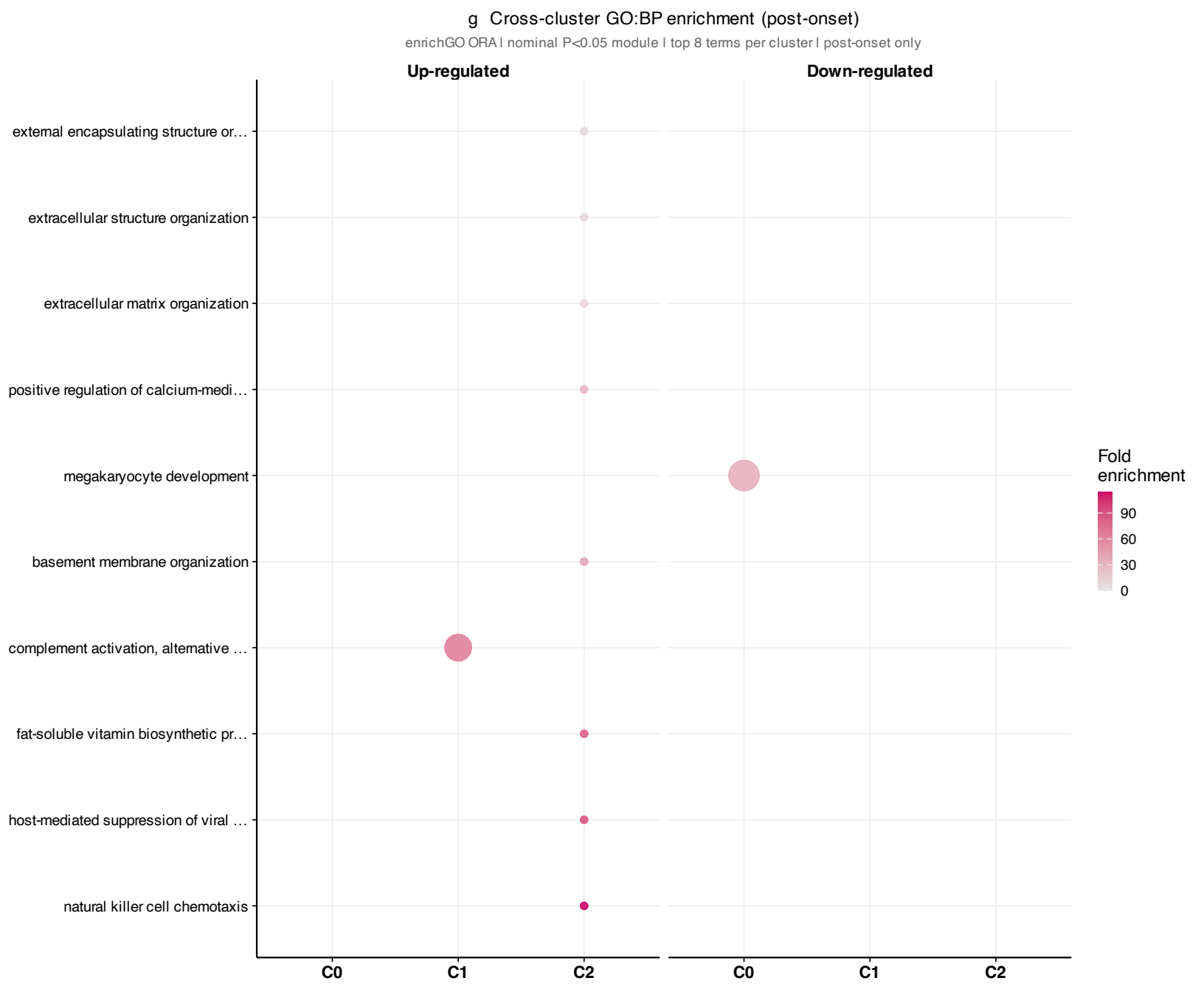

b C1 vs MS-None · cranial nerve / demyelinating (post-on
Up: 0 | Down: 0 (FDR<0.05, n=2915 proteins) | post-onset MS only
Bonferroni
HSD11B1
CCL21
NPTN
PDXDC1
KRT6C
APOF
ITGBL1
PZP
UHRF2
SERPINA11
$-\log_{10}(P)$
$\log_2$ fold change (cluster vs MS-None)
5
4
3
2
1
0
-0.5
0.0
0.5
1.0

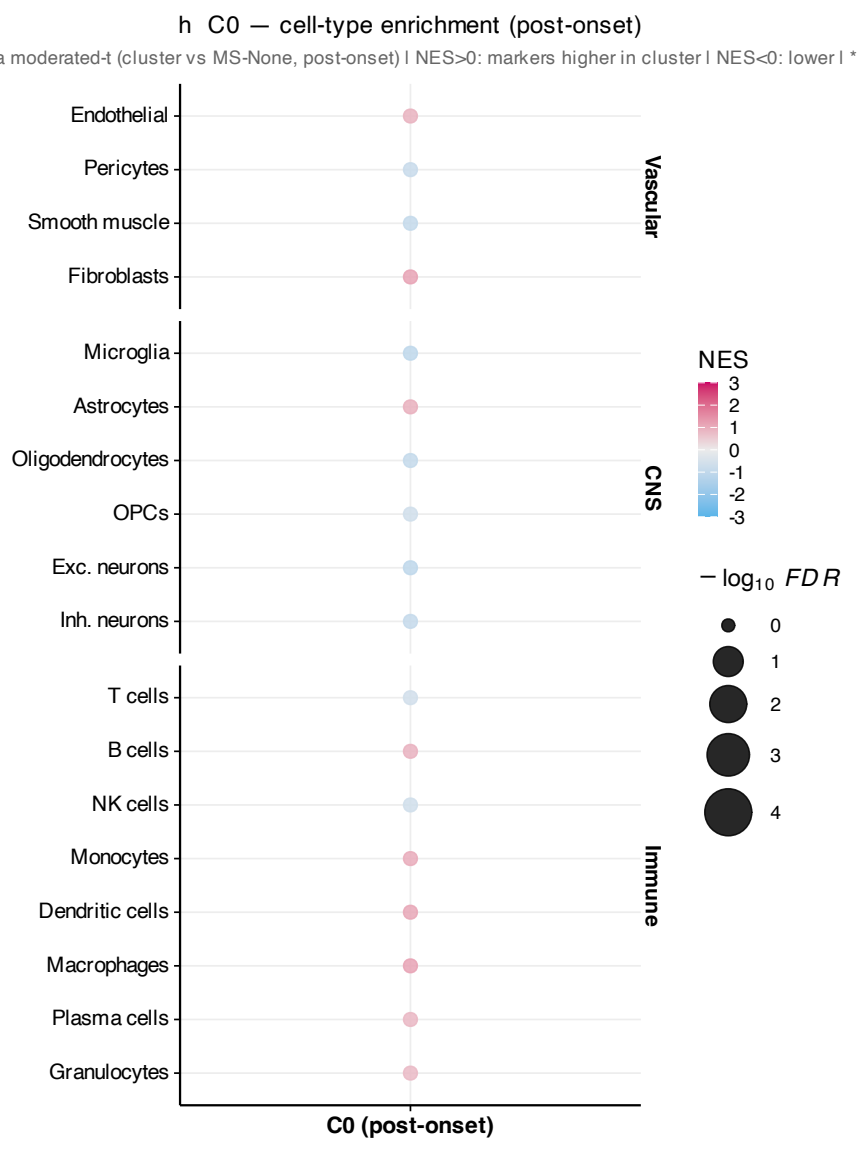

c C2 vs MS-None · neurological sx + EBV (post-onset
Up: 0 | Down: 0 (FDR<0.05, n=2915 proteins) | post-onset MS only
Bonferroni
CCL3
MMP7
NGRN
PKD2
CDHR5
FLT1
NPHS2
HIP1R
HJV
CCL4
$-\log_{10}(P)$
$\log_2$ fold change (cluster vs MS-None)
5
4
3
2
1
0
-2
-1
0
1

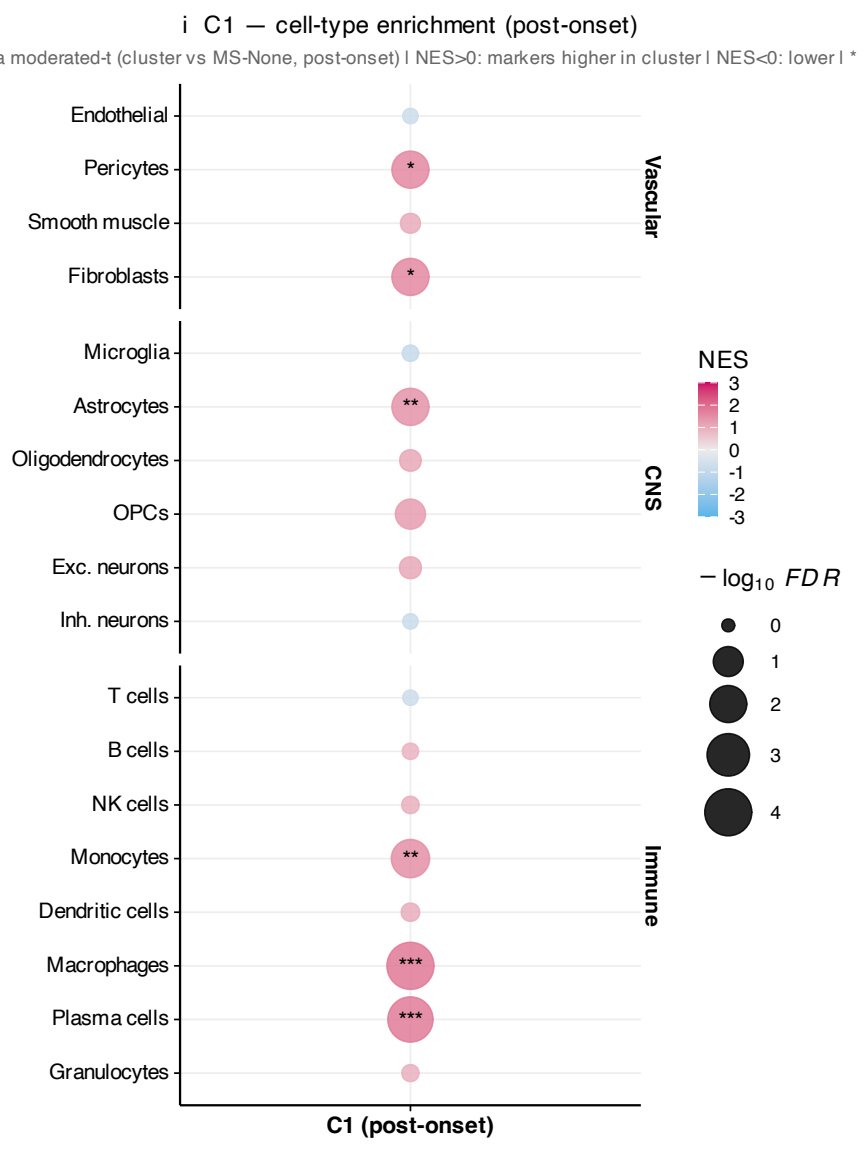

d C0 — GO:BP enrichment (post-onset)
enrichGO ORA | nominal P<0.05 module vs MS-None | post-onset only
megakaryocyte development
Down in C0 · post-onset
0
10
20
30
Fold enrichment
Gene count
4
direction
Down in C0 · post-onset

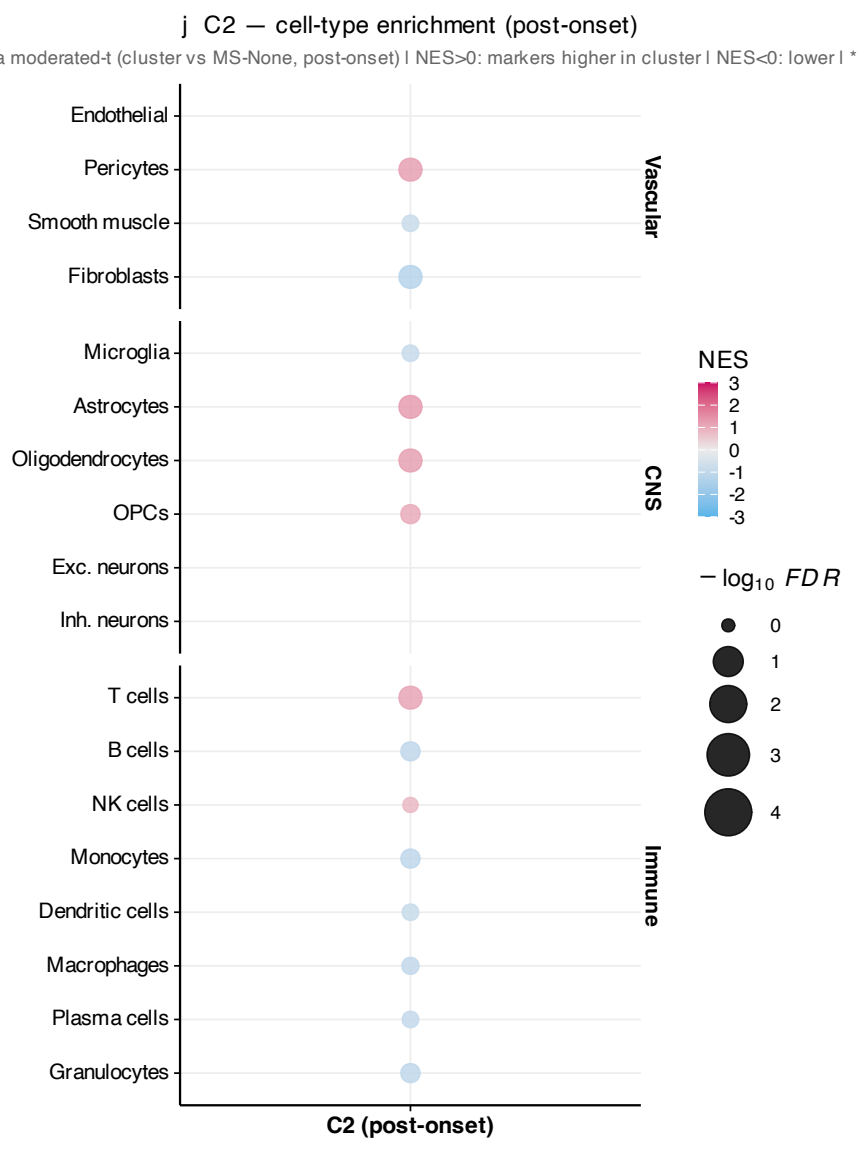

e C1 — GO:BP enrichment (post-onset)
enrichGO ORA | nominal P<0.05 module vs MS-None | post-onset only
complement activation, alternative ...
Up in C1 (cranial/demyel.) · post-onset
0
20
40
60
Fold enrichment
Gene count
3
direction
Up in C1 (cranial/demyel.) · post-onset

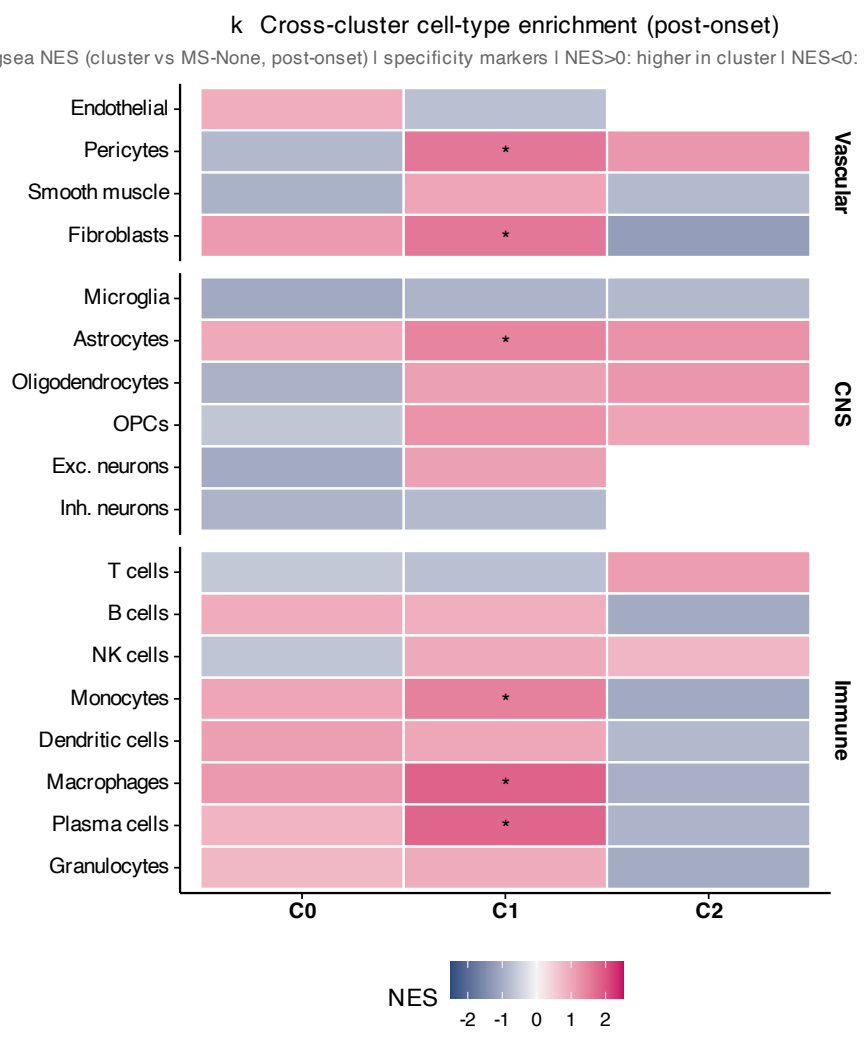

f C2 — GO:BP enrichment (post-onset)
enrichGO ORA | nominal P<0.05 module vs MS-None | post-onset only
natural killer cell chemotaxis
host-mediated suppression of viral …
fat-soluble vitamin biosynthetic pr…
host-mediated suppression of viral …
positive regulation of lymphocyte c…
basement membrane organization
positive regulation of calcium-medi…
extracellular matrix organization
extracellular structure organization
external encapsulating structure or…
0
30
60
90
120
Fold enrichment
Up in C2 (neurological/EBV) · post-onset
direction
Up in C2 (neurological/EBV) · post-onset
Gene count
2
3
4
5

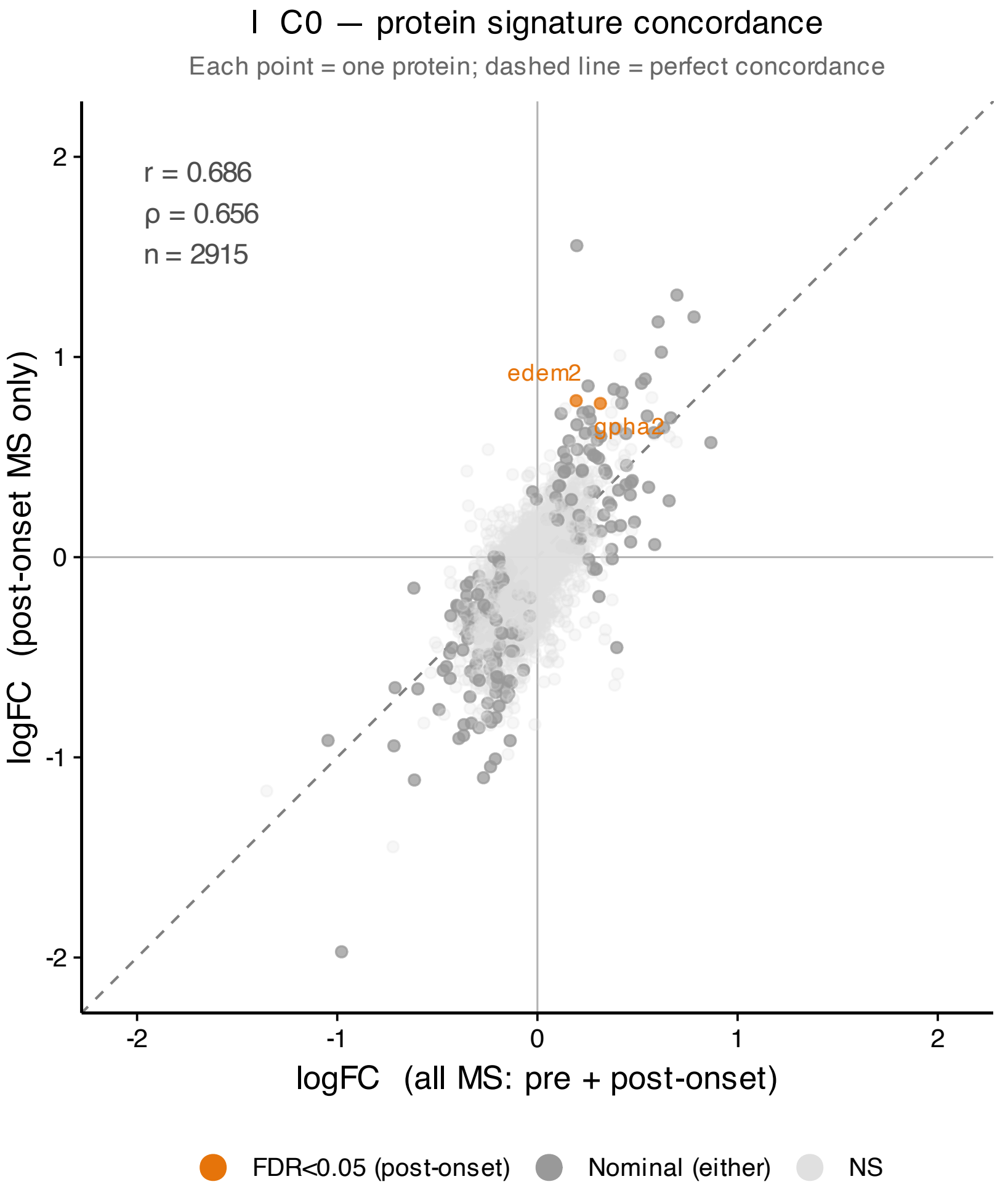

g Cross-cluster GO:BP enrichment (post-onset)
enrichGO ORA | nominal P<0.05 module | top 8 terms per cluster | post-onset only
Up-regulated
Down-regulated
external encapsulating structure or…
extracellular structure organization
extracellular matrix organization
positive regulation of calcium-medi…
megakaryocyte development
basement membrane organization
complement activation, alternative …
fat-soluble vitamin biosynthetic pr…
host-mediated suppression of viral …
natural killer cell chemotaxis
C0
C1
C2
C0
C1
C2
Fold enrichment
90
60
30
0

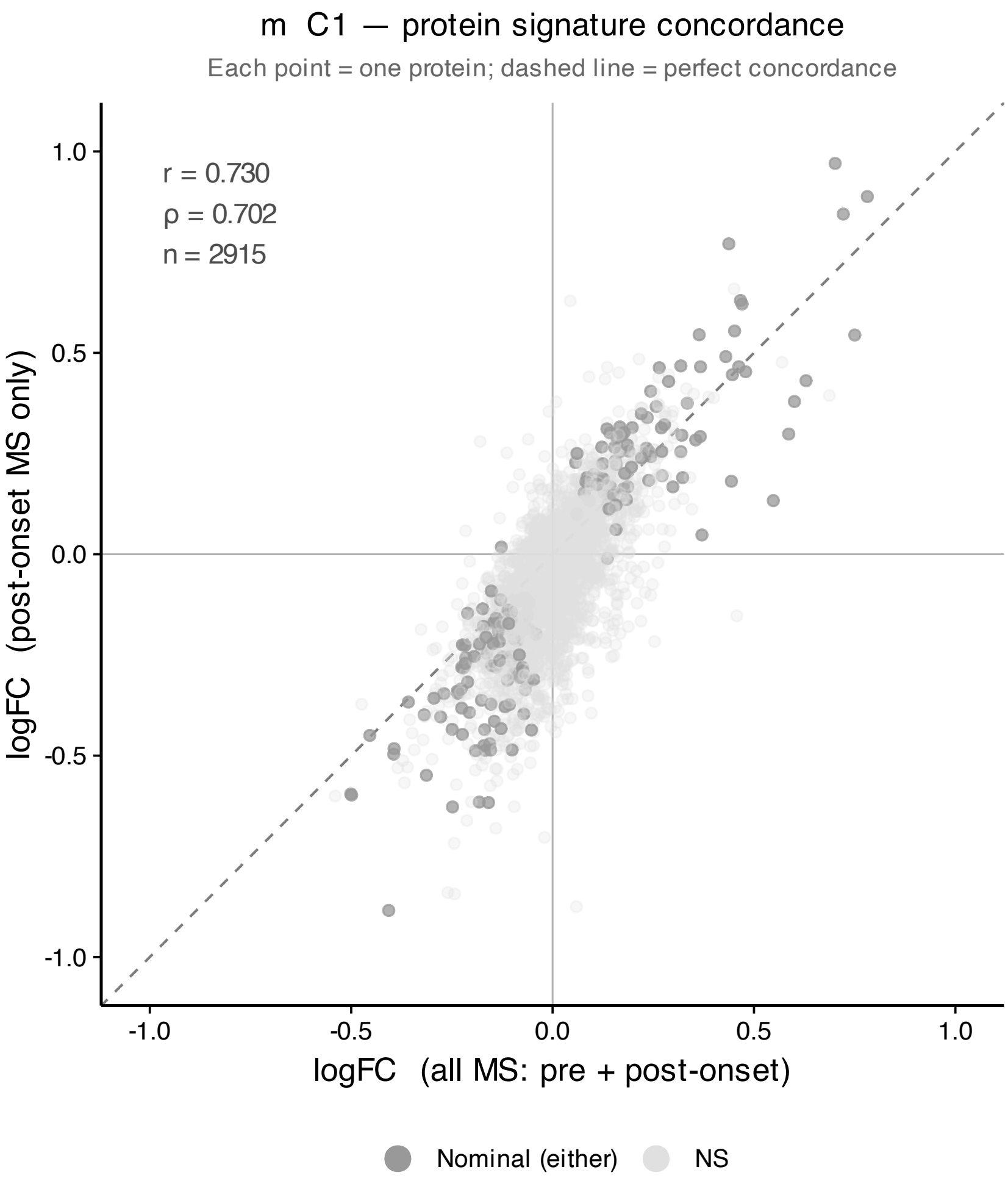

h C0 — cell-type enrichment (post-onset)
a moderated-t (cluster vs MS-None, post-onset) | NES>0: markers higher in cluster | NES<0: lower | *
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Vascular
Microglia
Astrocytes
Oligodendrocytes
OPCs
Exc. neurons
Inh. neurons
CNS
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
Immune
C0 (post-onset)
NES
3
2
1
0
-1
-2
-3
− log10 FDR
0
1
2
3
4

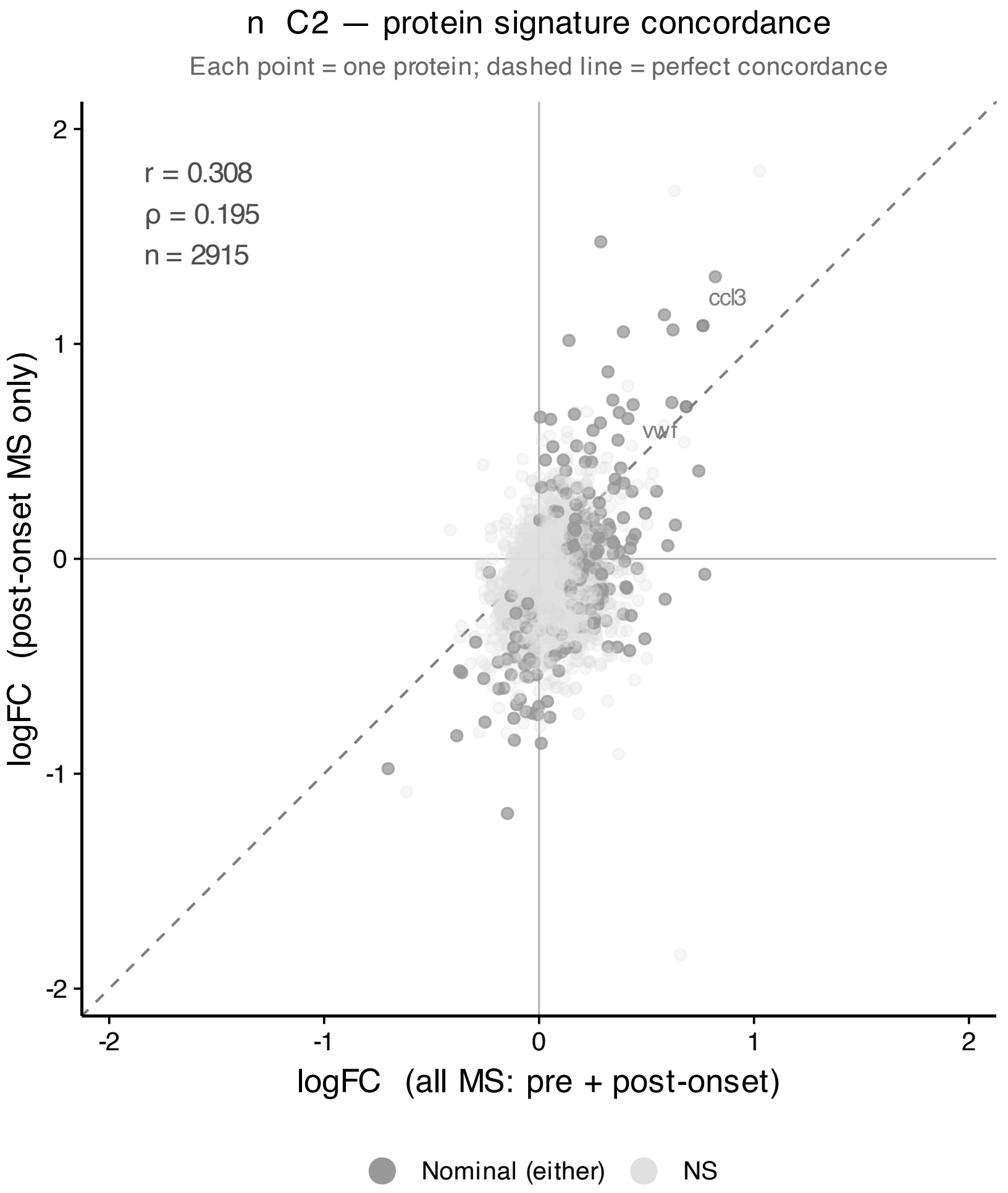

i C1 — cell-type enrichment (post-onset)
a moderated-t (cluster vs MS-None, post-onset) | NES>0: markers higher in cluster | NES<0: lower | *
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Microglia
Astrocytes
Oligodendrocytes
OPCs
Exc. neurons
Inh. neurons
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
Vascular
CNS
Immune
C1 (post-onset)
NES
3
2
1
0
-1
-2
-3
− log10 FDR
0
1
2
3
4


j C2 — cell-type enrichment (post-onset)
a moderated-t (cluster vs MS-None, post-onset) | NES>0: markers higher in cluster | NES<0: lower | *
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Vascular
Microglia
Astrocytes
Oligodendrocytes
OPCs
Exc. neurons
Inh. neurons
CNS
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
Immune
C2 (post-onset)
NES
3
2
1
0
-1
-2
-3
− log10 FDR
0
1
2
3
4


k Cross-cluster cell-type enrichment (post-onset)
gsea NES (cluster vs MS-None, post-onset) | specificity markers | NES>0: higher in cluster | NES<0:
Endothelial
Pericytes
Smooth muscle
Fibroblasts
Vascular
Microglia
Astrocytes
Oligodendrocytes
OPCs
Exc. neurons
Inh. neurons
CNS
T cells
B cells
NK cells
Monocytes
Dendritic cells
Macrophages
Plasma cells
Granulocytes
Immune
C0
C1
C2
NES
-2
-1
0
1
2


I C0 — protein signature concordance
Each point = one protein; dashed line = perfect concordance
r = 0.686
ρ = 0.656
n = 2915
edem2
gpha2
logFC (post-onset MS only)
logFC (all MS: pre + post-onset)
2
1
0
-1
-2
-2
-1
0
1
2
FDR<0.05 (post-onset)
Nominal (either)
NS


m C1 — protein signature concordance
Each point = one protein; dashed line = perfect concordance
r = 0.730
ρ = 0.702
n = 2915
1.0
0.5
0.0
-0.5
-1.0
logFC (post-onset MS only)
-1.0
-0.5
0.0
0.5
1.0
logFC (all MS: pre + post-onset)
Nominal (either)
NS


n C2 — protein signature concordance
Each point = one protein; dashed line = perfect concordance
r = 0.308
ρ = 0.195
n = 2915
ccl3
vwf
logFC (post-onset MS only)
logFC (all MS: pre + post-onset)
2
1
0
-1
-2
-2
-1
0
1
2
Nominal (either)
NS